FILE PERMISSIONS IN LINUX

What are File permissions

The file permissions are the permissions that assign to every file, folder or anything in the system. That defines:

- **Who can access the file** (user, group, others)
- **What actions they can perform** (read, write, execute)

1.mkdir dir1

Prompt: {d rwxr-xr-x 2 root root 4096 Apr 11 10:25 dir1}

1. d ----------> file type
2. rwx ----------> permissions (owner/groups/others)
3. 2 ----------> link count
4. root ----------> ownership (owner/group)
5. 4096 ----------> size
6. April 11 ----------> date & time
7. dir1 ----------> file name <directory>

1.FILE TYPE

Users define files

1.normal file (-)
2.directory (d)
3.link file (l)

System defines files

4.socket device file (s)
5.character device file (c)
6.pipe file (p)
7.block device file (b)

Note:

ls -l /dev | head ----> To check the system files

2.PERMISSIONS (users/groups/other)

We can assign a permission to a user (owner), group or other users

1. User (Owner)

- The person who created the file
- Has special control over the file
- Example: If user1 creates a file, then user1 is the owner

2. Group

- A collection of users
- All users in the same group can share permissions
- Example: if dev team group, so all developers can access the file

3. Other

- Everyone else in the system
- Users who are not owner and not in the group

Types Of Permissions

- **r → Read (4)** (open/view file) ls (long listing)
- **w → Write (2)** (edit/delete file)
- **x → Execute (1)** (run file like a program) change directory, command, script

In Binary form: -

- rwx-------> 111
- r-x--------> 101
- rw--------> 110
- r-- --------> 100
- -w- -------> 010
- --x --------> 001

Example:

In Binary Form		In octal form	
rwx	111	$1\times2^2 + 1\times2^1 + 1\times2^0 = 4 + 2 + 1$	7
rw-	110	$1\times2^2 + 1\times2^1 + 0\times2^0 = 4 + 2 + 0$	6
r-x	101	$1\times2^2 + 0\times2^1 + 1\times2^0 = 4 + 0 + 1$	5
r--	100	$1\times2^2 + 0\times2^1 + 0\times2^0 = 4 + 0 + 0$	4
-wx	011	$0\times2^2 + 1\times2^1 + 1\times2^0 = 0 + 2 + 1$	3
-w-	010	$0\times2^2 + 1\times2^1 + 0\times2^0 = 0 + 2 + 0$	2
--x	001	$0\times2^2 + 0\times2^1 + 1\times2^0 = 0 + 0 + 1$	1
---	000	$0\times2^2 + 0\times2^1 + 0\times2^0 = 0 + 0 + 0$	0

Default permissions:

For Root User:

- file ----> 644 (rw- r-- r--)
- dir ----->755 (rwx r-x r-x)

For Local User:

- file ----> 664 (rw- rw- r--)
- dir ----> 775 (rwx rwx r-x)

CLOUDBLITZ

To change the permissions:

- **u---------> user (owner)**
- **g---------> group**
- **o---------> other**

(+) ----------> add the permission
(-) ----------> remove the permission
(=) ----------> replace & set the permission

1st method (symbolic):

1.chmod u+x file1
rwx --- ---

2.chmod u-x file1
rw- --- ---

3.chmod ugo=rwx file1
rwx rwx rwx

2nd method (octal):

1.chmod 777 file1
rwx rwx rwx

2.chmod 644 file1
rw-r--r—

3.chmod 755 file1
rwx r-x r-x

CLOUDBLITZ

How to read the files permissions
1.ls -l file

How to read the directory permissions
1.ls -ld dir

Umask

Umask giving default value to a particular root user or local user when creates a file or directory

Root:
Umask – 022

Local user:
Umask – 002

Full permissions

1. File ----------> 666
2. Directory -----> 777

How to calculate permissions for root user

Umask = full permissions – default permissions
= 666 – 644
= 022

How to calculate permissions for local user

Umask = full permissions – default permissions
= 666 – 664
= 002
